

4.3.7 It suffices to prove AX , NM and ED are concurrent

Angle since $\angle CNM = \angle ANM$ (M is the arc mid-pt),

Bisector Therefore it suffices to prove (by Ceva),

$$\frac{\sin \angle NAX}{\sin \angle EAX} \cdot \frac{\sin \angle DEA}{\sin \angle DEN} = 1 \quad \text{--- (1)}$$

$$\frac{\sin \angle NAX}{\sin \angle EAX} \cdot \frac{\sin \angle DEN}{\sin \angle DEA} = 1$$

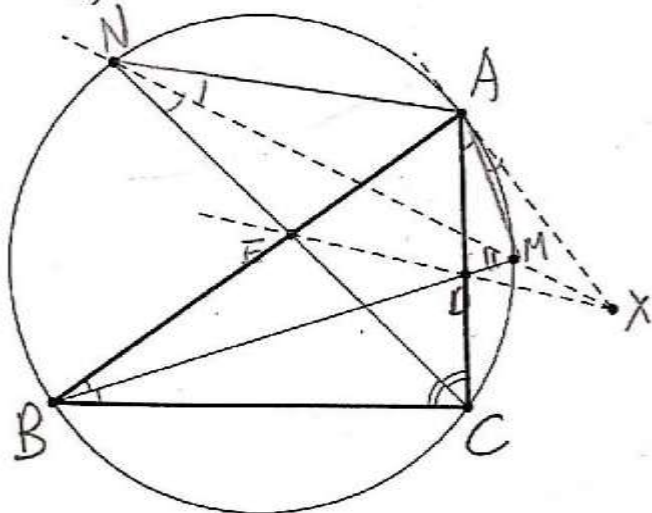
$$\text{We have } \frac{\sin \angle NAX}{\sin \angle EAX} = \frac{\sin \frac{C}{2}}{\sin \frac{C}{2}}$$

$$\text{and } \frac{\sin \angle DEA}{\sin \angle DEN} = \frac{\sin \angle DEA}{\sin \angle EDA} \cdot \frac{\sin \angle EDX}{\sin \angle DEC} = \frac{DA}{EA} \cdot \frac{EC}{DC} = \frac{AB}{BC} \cdot \frac{\sin A}{\sin \frac{C}{2}} \cdot \frac{\sin C}{\sin A} = \frac{\sin C}{\sin \frac{C}{2}}$$

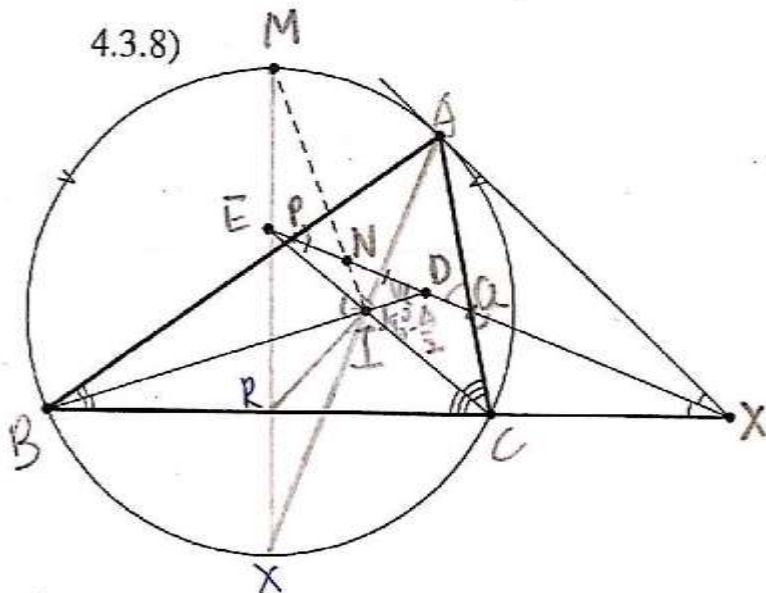
This establishes (1).

Scanned with CamScanner

4.3.7)



4.3.8)



Scanned with CamScanner

4.3.8

Angle
Bisector
CQWe have $\angle AXB = \angle BCA - \angle XAC = \angle BCA - \angle ABC = \frac{C-B}{2}$

$$\angle DQC = \angle QXC + \angle ACX = 180^\circ - C + \frac{C-B}{2} = 180^\circ - \frac{C+B}{2} = 90^\circ + \frac{A}{2} = \angle BIC$$

 $\therefore \triangle IBC$ is cyclic. $\therefore \angle EDS = \angle ICA = \angle ECB$ $\therefore \triangle ECB$ is cyclic.Let R be the midpt of BC , then since $\triangle EID \sim \triangle BIC$,
 $\triangle ENI \sim \triangle BRI \Rightarrow \angle EIN = \angle BRI$ Meanwhile, since X is the centre of (BIC) and $\angle MBX = \angle MCX = 90^\circ$,
 MB, MC are tangent to (BIC) , so MI is the symmedian of $\triangle BIC$

$$\therefore \angle MIE = \angle BRI = \angle EIN$$

 $\therefore M, N, I$ are collinear.